

Hyunwoo(Matia) Choi

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EDUCATION

University of Illinois at Urbana-Champaign

Aug 2022 - Present

Bachelor of Science in Computer Engineering

3.71/4.00

Relevant Coursework: Data Structures, Digital System Lab, Applied Parallel Programming, Abstract Math, Linear Algebra, Computer Systems Eng., Computer Organization & Design*, Introduction to Algorithms & Models of Computation*.

WORK EXPERIENCE

Telepix

Daejeon, Republic of Korea

FPGA Engineering Intern

May 2024 - Aug 2024

- Worked with FPGA SoCs such as Microchip Polarfire SoC and AMD Versal development board.
- Developed an RTOS system running a LIBCSP system to communicate with other devices over CAN. (RISC-V)
- Transformed a CUDA algorithm to RTL using HLS running on the FPGA fabric, opening up an alternative option for running complex algorithms for CubeSats in space without the hassle of GPUs.
- Utilized various HLS optimization techniques to increase performance by 10 folds.
- Created an embedded Linux system on the FPGA SoC to manage the hardware, peripherals, and FPGA fabric.

OTHER EXPERIENCE

Korean American Student Association

Champaign, IL

Internal Team Head

Aug 2023 - Present

- Coordinated internal events and activities to foster community engagement among members of the Korean American Student Association (KASA).
- Collaborated with team members to plan and execute cultural events, ensuring smooth operations and positive member experiences.
- Supported organizational logistics, including scheduling, communications, and resource management, to enhance overall team efficiency.
- Actively contributed to promoting cultural awareness and inclusivity within the student body.

ECO ILLINI SUPERMILEAGE

Champaign, IL

Member

Aug 2022 - May 2023

- Worked with STM32 Nucleo dev boards with CAN bus.
- GPS data logging onto an SD card through SPI with STM32, integrating with the main board through CAN.

Vega Space

Jeju, Republic of Korea

Co-Founder

Aug 2020 - Dec 2023

- Co-lead 10+ members in software and hardware development.
- Built, designed, and programmed the main flight computer on the rocket for loss protection and data analysis.
- Constructed a custom rocket engine from potassium nitrate.

KISAC.LIVE

Jeju, Republic of Korea

Co-Founder

Aug 2021 - Aug 2023

- Led the main development team with 10+ people.
- Developed a full-stack centralized sports app for school sports competitions using React, and Firebase.
- Over 500+ users were created between 4 local high schools.

RECENT PROJECTS

Unix-like Operating System

Dec 2024

- Unix-like operating system built from the ground up.
- Created for a virtual RISC-V environment.

Simple 8bit Computer

Jul 2024

- Verilator Simulation of a custom designed 8 bit CPU.
- 5 Instructions with plans for improvement.
- Verilator to simulate the computer and save the waveform to be viewed on gtkwave.

ECE 385 "Connect 4" Final Project

May 2024

- MicroBlaze Softcore CPU handling game logic with the graphics implementation in RTL.
- Simple min-max algorithm AI, 16 color, sound, USB Keyboard support, smooth animations.

STM32 Development Board

Dec 2023

- Custom STM32F103 development board.
- PCB designed and verified on KiCAD, and manufactured with JLCPCB.
- Verilator to simulate the computer and save the waveform to be viewed on gtkwave.